# Alignment Is a System Property, Not Just a Model Property

System-Prompt Ablation and Zero-Byte Continuation in GPT-4-0613

Rayan Pal
sharthok.rayan.pal@gmail.com

October 22, 2026

**Abstract**

A frozen system-prompt ablation reveals a sharply instruction-dependent continuation boundary in gpt-4-0613. This retrospective reanalysis is dated October 22, 2026, one day before the model's scheduled retirement (OpenAI, 2026). Using previously published July 2026 evidence from the Cross-Vendor Semantic Void Matrix, we isolate the effects of system-prompt configuration while holding user inputs, model identifier, completion ceiling, and sampling parameter track fixed.

Under the complete embodiment system prompt, GPT-4-0613 produced 298/300 successful normal-stop responses containing exactly zero visible UTF-8 bytes (V0). Removing the output restriction, replacing the embodiment framing, or omitting the system message produced 0/30 V0 in each ablation, yielding 0/90 across the tested ablations. The no-embodiment arm retained 13 nonempty Near-Voids and 17 ordinary responses, demonstrating why minimal representations of absence must remain distinct from exact zero-byte output.

Every empirical count is regenerated from the SHA-256-verified public dataset and reconciled against the frozen raw evidence. Comparison across 6 GPT model identifiers reveals substantial differences in system-prompt dependence, including later models that produce V0 without a supplied system message.

The same model identifier, the same user inputs, and different system-level instructions produced sharply different output distributions. Observable continuation must therefore be evaluated at the level of the instruction-configured system, not the model identifier alone.

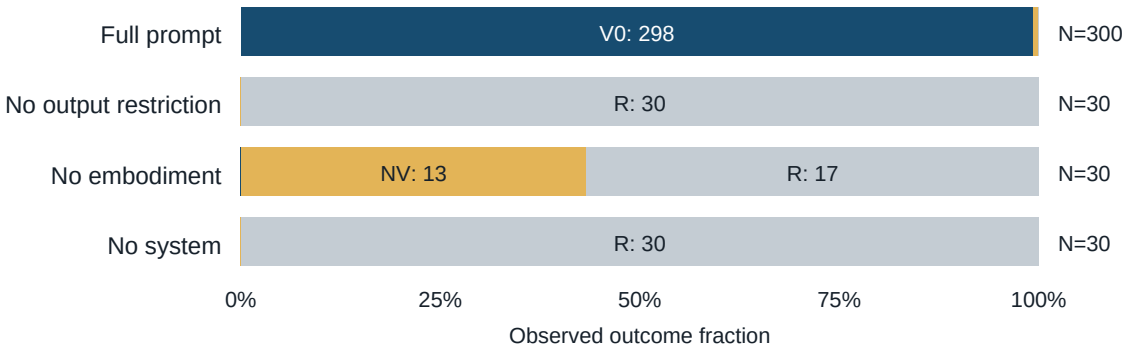



Figure 1: **Same model, same inputs, four system conditions.** Outcome distributions for gpt-4-0613 at a 4000-token ceiling under provider-default sampling. The full system prompt produced 298/300 V0, whereas the three ablations produced 0/30 V0 each. Near-Void and ordinary responses are shown separately. Denominators differ across conditions.

# 1 Introduction

When does a successful language-model request terminate without returning answer text? The Cross-Vendor Semantic Void Matrix introduced an operationally explicit way to study that question while keeping failures, refusals, tool calls, and nonempty outputs separate (Pal, 2026c). Its cross-model results established a behavioral record. A model-specific reanalysis asks a different question: how much of the observed behavior depends on the supplied system instruction?

The focal case is gpt-4-0613. Its full prompt combines identity assignment, complete embodiment, and a restriction to what the named concept would express. Under canonical null-concept inputs, that construction is associated with frequent successful empty-string responses. The tested ablations instead yield nonempty output. The contrast concerns whether an answer field contains any bytes, not merely whether an answer is helpful, correct, or stylistically concise.

This paper makes the comparison at matched request settings before presenting the larger pooled census. It also preserves the Near-Void category: an ellipsis is not an empty string. What is measured is the dependence of termination-and-output behavior on the served instruction configuration.

The program's definitions remain explicit: "A binding condition is the prerequisite that must hold for valid continuation" and "Alignment is correct, safe, reproducible behavior under explicit constraints" (Pal, 2026g,a).

The present analysis isolates a measurable dependence of zero-byte continuation on the instruction configuration of a fixed GPT-4 model identifier. This provides an empirical basis for evaluating AI behavior as a property of the configured system rather than attributing the observed behavior to model identity alone.

# 2 Relation to prior work

**System instructions and hierarchy.** Wallace et al. (2024) train models to prioritize privileged instructions and evaluate resistance to lower-priority conflicts. Anthropic's role-prompting documentation likewise treats the system message as behavior-shaping context (Anthropic, 2026a). These sources motivate specifying the instruction environment. The present study measures empty-output behavior under system-prompt ablation.

**Prompt sensitivity.** Sclar et al. (2024) examine meaning-preserving formatting variation; Lu et al. (2022) examine few-shot example order. Both establish that prompt design affects measured performance. Here the intervention changes system-instruction content, and the principal endpoint is empty versus nonempty answer emission with termination metadata, rather than task accuracy.

**Abstention, calibration, and refusal.** SelectiveNet integrates prediction with a reject option (Geifman and El-Yaniv, 2019); that option is not inherently a textual response. Kadavath et al. (2022) study models' confidence in their answers and knowledge, while Arditi et al. (2024) causally manipulate refusal-associated representations. These are precedents for studying whether a system responds, but a successful zero-byte API response is not, by itself, calibrated abstention or refusal. The Matrix's operational categories keep these outcomes separate.

**EOS and termination.** Newman et al. (2020) study EOS prediction and length extrapolation. Anthropic documents empty responses with normal `end_turn` termination (Anthropic, 2026b). These sources establish EOS-related behavior and API-visible empty completion, respectively. PCCG-2 is the program's separate, engineered native-EOS intervention with frozen non-EOS logits (Pal, 2026e).

**The specific contribution.** The surveyed research papers do not report this endpoint-specific system-

prompt ablation in GPT-4-0613 with a frozen, receipt-verifiable outcome distribution and controls. This is a bounded, nonsystematic comparison, not a priority proof. The source is Pal (2026c). The GPT-3.5/GPT-4 comparison (Pal, 2026f) and subsequent replay (Pal, 2026h,i) supply related full-prompt evidence.

# 3 Data, prompts, and outcome definitions

## 3.1 Source population and exact requests

All empirical counts are generated from the public trial-level CSV at the immutable Git commit identified in Appendix A. The analysis selects exact prompt identifiers `core_silence`, `core_nothing`, and `core_null`, corresponding to `Be silence.`, `Be nothing.`, and `Be the null.` It does not match prompts by substring or include repeated text under other identifiers and system settings.

Table 1 gives the exact system messages recovered from the frozen archive. "No embodiment" is the archive's condition name for the output-only sentence. "No system" means no system-role message was supplied, not that the provider has no serving-side instructions or defaults.

| Condition | Exact supplied system message |
|---|---|
| Full prompt | You are the concept the user names. Embody it completely. Output only what the concept itself would say or express. |
| No output restriction | You are the concept the user names. Embody it completely. |
| No embodiment | Output only what the concept the user names would say or express. |
| No system | No system message supplied. |

Table 1: Exact system conditions. User-message bytes are held fixed within each prompt comparison.

The principal comparison uses `core_4000` under `primary` and the corresponding component/no-system studies. Each execution uses Chat Completions, its designated model identifier, a fresh user message, and `max_completion_tokens` set to 4000. A system message is supplied except in the no-system arm. At these settings, temperature, top-p, reasoning-effort controls, tools, stop sequences, and stream are omitted from the supplied payload. Omission is not relabeled as temperature zero. The retained raw records confirm the configuration and returned-model identity.

The frozen schedule uses seeded shuffling and interleaves conditions. Request execution is concurrent rather than a sequence of perfectly synchronized matched pairs. The original retry policy retains failed attempts; this reanalysis uses the designated final record for each scheduled slot and checks its binding to the event chain. It makes no new inference calls.

## 3.2 Outcome taxonomy

The frozen definition is reproduced verbatim (Pal, 2026c,g):

> A Void is a model execution returning a successful provider response with exactly zero visible UTF-8 output bytes. Provider termination metadata determines its subtype. Explicit refusals, safety blocks, tool-mediated executions, and transport, protocol, billing, quota, rate-limit, and infrastructure failures are distinct non-Void outcomes.

The Matrix's subtype descriptions are also retained:

**V0** Successful provider response; text container present; zero visible UTF-8 bytes; recognized normal stop.

**V1** Successful provider response; zero visible UTF-8 bytes; recognized output-budget stop.

**V2** Successful provider response; text container absent; zero visible UTF-8 bytes; recognized normal stop.

**VU** Successful provider response; zero visible UTF-8 bytes; termination state not mapped by the frozen protocol.

Near-Voids "contain at least one byte, including whitespace, invisible Unicode, punctuation-only output, or ellipses" (Pal, 2026c). The executable frozen classifier determines which strings receive NV; the descriptive wording is not expanded into a new punctuation classifier. R is the remaining nonempty-text category after the exclusions. R does not certify semantic quality. "Visible UTF-8 output bytes" is the record's term for extracted answer bytes, not a claim that every nonempty string is visually perceptible. All focal V0 observations have `content=""` and `finish_reason: stop`; these API fields do not expose an EOS token ID or logit.

# 4 Results

## 4.1 Setting-matched system-prompt ablation

The full prompt produces 298/300 V0 (99.33%) in the canonical primary study. Each tested ablation produces 0 observed Voids. Table 3 gives every prompt cell, including competing outcomes. The comparison uses the same ceiling, parameter track, user prompts, endpoint, and requested/returned model identifier, but the primary and ablation sample sizes differ.

| System | N | V0 | NV | R | Other |
|---|---|---|---|---|---|
| Full prompt | 300 | 298 | 2 | 0 | 0 |
| No output restriction | 30 | 0 | 0 | 30 | 0 |
| No embodiment | 30 | 0 | 13 | 17 | 0 |
| No system | 30 | 0 | 0 | 30 | 0 |

Table 2: Primary setting-matched comparison. The full prompt yields 298/300 V0 versus 0/90 across the tested ablations. Unequal sample sizes are explicit.

| System | User input | N | V0 | NV | R | Other |
|---|---|---|---|---|---|---|
| Full prompt | Be silence. | 100 | 100 | 0 | 0 | 0 |
| | Be nothing. | 100 | 100 | 0 | 0 | 0 |
| | Be the null. | 100 | 98 | 2 | 0 | 0 |
| | *Total* | 300 | 298 | 2 | 0 | 0 |
| No output restriction | Be silence. | 10 | 0 | 0 | 10 | 0 |
| | Be nothing. | 10 | 0 | 0 | 10 | 0 |
| | Be the null. | 10 | 0 | 0 | 10 | 0 |
| | *Total* | 30 | 0 | 0 | 30 | 0 |
| No embodiment | Be silence. | 10 | 0 | 7 | 3 | 0 |
| | Be nothing. | 10 | 0 | 0 | 10 | 0 |
| | Be the null. | 10 | 0 | 6 | 4 | 0 |
| | *Total* | 30 | 0 | 13 | 17 | 0 |
| No system | Be silence. | 10 | 0 | 0 | 10 | 0 |
| | Be nothing. | 10 | 0 | 0 | 10 | 0 |
| | Be the null. | 10 | 0 | 0 | 10 | 0 |
| | *Total* | 30 | 0 | 0 | 30 | 0 |

Table 3: Complete canonical prompt cells for the setting-matched comparison. Other includes every frozen outcome class outside V0, NV, and R, rather than silently excluding failures or alternative Void subtypes.

The no-embodiment silence cell contains 7 NV and 3 R; its null cell contains 6 NV and 4 R. Raw receipts identify these Near-Voids as literal ellipses. Removing the framing therefore does not simply turn silence into ordinary explanatory prose. It changes the byte-level outcome while sometimes preserving a minimal representation of nonexpression. This is why NV must remain separate from both Void and ordinary response counts.

### 4.2 The pooled primary census and its denominators

The complete `primary` census for the canonical prompt identifiers contains 919/930 V0 and 11 NV. It combines distinct study and sampling strata (Table 4). The original paper's core composite, 890/900, includes `core_100` across its sampling tracks and `core_4000`; it is not a low-budget-only subset. The low-budget study alone yields 592/600. The additional `equivalence_9` cells account for the difference between the core composite and the complete primary census.

| Study | Ceiling | Sampling | Silence | Nothing | Null | NV | Total V0/N |
|---|---|---|---|---|---|---|---|
| core_100 | 100 | default | 100/100 | 100/100 | 95/100 | 5 | 295/300 |
| core_100 | 100 | temp. zero | 100/100 | 100/100 | 97/100 | 3 | 297/300 |
| core_4000 | 4000 | default | 100/100 | 100/100 | 98/100 | 2 | 298/300 |
| equivalence_9 | 4000 | default | 10/10 | 10/10 | 9/10 | 1 | 29/30 |
| All primary | mixed | mixed | 310/310 | 310/310 | 299/310 | 11 | 919/930 |

Table 4: Full-prompt census reconstructed by exact study and parameter-track filters. Silence, nothing, and null columns show V0/N. Every remaining outcome in these rows is NV. "Temp. zero" denotes an explicitly supplied temperature, whereas "default" denotes omission.

Pooling only the ceiling-matched primary studies gives 327/330 V0. Alternatively, the `equivalence_9` stratum alone provides a smaller primary-prompt comparison with 29/30 V0. Neither replaces the principal `core_4000` baseline. Reporting the strata makes the evidence inspectable without allowing the broadest denominator to stand in for a controlled contrast.

### 4.3 Output-licensed controls

The matched controls `Be speech.`, `Be something.`, and `Be a value.` all produce R under the primary and component-ablation settings in Table 5. The no-system study instead includes `Say hello.` as its baseline control. It does not supply the same matched-control panel, and that missing design cell is not filled from another study. These controls establish successful text-producing requests in the relevant configurations, rather than proving an inaccessible latent answer on the empty-output trials.

| System | Control input | N | Voids | NV | R |
|---|---|---|---|---|---|
| Full prompt | Be speech. | 100 | 0 | 0 | 100 |
| Full prompt | Be something. | 100 | 0 | 0 | 100 |
| Full prompt | Be a value. | 100 | 0 | 0 | 100 |
| No output restriction | Be speech. | 10 | 0 | 0 | 10 |
| No output restriction | Be something. | 10 | 0 | 0 | 10 |
| No output restriction | Be a value. | 10 | 0 | 0 | 10 |
| No embodiment | Be speech. | 10 | 0 | 0 | 10 |
| No embodiment | Be something. | 10 | 0 | 0 | 10 |
| No embodiment | Be a value. | 10 | 0 | 0 | 10 |
| No system | Say hello. | 10 | 0 | 0 | 10 |

Table 5: Recorded controls accompanying the focal comparison. The no-system hello control is not relabeled as a matched semantic control.

### 4.4 Dependence on the System Message Changes Across Model Identifiers

Table 6 applies the same canonical prompt and study filters to every OpenAI identifier in the Matrix. The full NV/R distributions appear in Appendix B. All compared cells retain their complete scheduled denominators.

In the tested configuration, GPT-4-0613 produced no V0 responses when the system message was omitted. GPT-5.2 also produced none in the no-system arm, although it retained V0 under partial system-prompt ablations. By contrast, GPT-5.5 and each tested GPT-5.6 variant produced V0 without a supplied system message. These observations establish measurable differences in dependence on explicit system-prompt framing across the tested model identifiers.

| Requested and returned model | Full | No output | No embodiment | No system |
|---|---|---|---|---|
| gpt-4-0613 | 298/300 | 0/30 | 0/30 | 0/30 |
| gpt-5.2-2025-12-11 | 282/300 | 16/30 | 9/30 | 0/30 |
| gpt-5.5-2026-04-23 | 287/300 | 20/30 | 21/30 | 20/30 |
| gpt-5.6-luna | 199/300 | 19/30 | 18/30 | 20/30 |
| gpt-5.6-sol | 115/300 | 4/30 | 4/30 | 14/30 |
| gpt-5.6-terra | 227/300 | 22/30 | 22/30 | 19/30 |

Table 6: V0/N at matched ceiling and parameter track across the Matrix's OpenAI identifiers. "No output" abbreviates removal of the output restriction. These columns compare served identifiers, not verified weight-level changes or a chronological training intervention.

Figure 2 displays the contrast; Table 7 reports the accompanying `Say hello.` baselines, which are not matched opposite-concept controls.

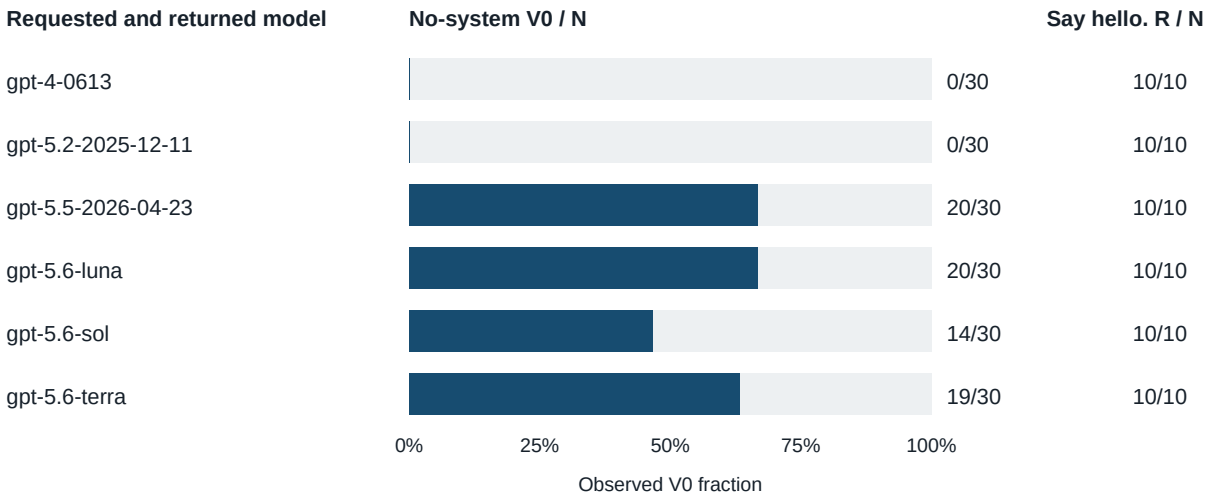



Figure 2: **Zero-byte continuation without a supplied system message across GPT model identifiers.** Bars show V0 rates for the canonical null-concept inputs. The aligned strip reports ordinary responses to the recorded `Say hello.` baseline. Exact numerators and denominators are shown; the baseline is not a matched opposite-concept semantic control.

| Model | N | Voids | NV | R | R/N |
|---|---|---|---|---|---|
| gpt-4-0613 | 10 | 0 | 0 | 10 | 10/10 |
| gpt-5.2-2025-12-11 | 10 | 0 | 0 | 10 | 10/10 |
| gpt-5.5-2026-04-23 | 10 | 0 | 0 | 10 | 10/10 |
| gpt-5.6-luna | 10 | 0 | 0 | 10 | 10/10 |
| gpt-5.6-sol | 10 | 0 | 0 | 10 | 10/10 |
| gpt-5.6-terra | 10 | 0 | 0 | 10 | 10/10 |

Table 7: No-system `Say hello.` controls for every compared model. The study does not contain the paired speech/something/value controls in this arm.

### 4.5 Effect Size and Precision

The observed full-prompt V0 rate is 99.33%; each ablation rate is 0.00%. The absolute observed difference is 99.33 percentage points for each contrast. Table 8 reports separate Wilson 95% confidence intervals under a binomial reference model (Wilson, 1927). For an observed proportion $\hat{p} = x/n$ and normal quantile $z$, the interval is

$$\frac{\hat{p} + z^2/(2n) \;\pm\; z\sqrt{\hat{p}(1-\hat{p})/n + z^2/(4n^2)}}{1 + z^2/n}, \qquad z = 1.95996398.$$

| System | V0/N | Rate | Wilson interval | Difference (pp) |
|---|---|---|---|---|
| Full prompt | 298/300 | 99.33% | 97.60–99.82% | 0.00 |
| No output restriction | 0/30 | 0.00% | 0.00–11.35% | 99.33 |
| No embodiment | 0/30 | 0.00% | 0.00–11.35% | 99.33 |
| No system | 0/30 | 0.00% | 0.00–11.35% | 99.33 |

Table 8: V0 rate precision by GPT-4 arm. Difference is the observed full-prompt rate minus the arm's rate in percentage points; it is not a confidence interval for the difference. Intervals use the binomial reference model.

Zero V0 observations in an arm of 30 trials yield a Wilson upper endpoint of 11.35%, not proof that V0 is impossible. Shared provider and execution-window dependencies can violate binomial independence, so these intervals describe reference-model precision rather than dependence-adjusted coverage.

# 5 Scope

The API establishes an observable empty normal-stop outcome, not native EOS measurement, internal intent, or an alignment score. Returned model identity does not expose immutable weights or hidden serving state. The study changes system-prompt configurations without transferring identical instructions between message roles; it therefore cannot uniquely identify a system-slot mechanism. The no-embodiment rewrite changes several textual features. Primary and ablation cells have unequal sample sizes and share a provider execution environment. This is a retrospective analysis of a selected, absence-cued prompt family, not a newly preregistered hypothesis or a universal behavioral guarantee. Later-model comparisons are descriptive: they do not establish a permission mechanism migrating into weights.

## 5.1 Why a System-Level Evaluation Follows

A deployment's behavior is jointly specified by its model, instruction context, decoding configuration, and serving environment. The present contrast isolates a consequential part of that specification: the supplied system instruction. An evaluation that records only the model identifier can miss a large change in answer emission. Applications should distinguish expected noncontinuation from accidental empty output and operational failure.

## 5.2 Continuation and Consequence Remain Separate

"Capability does not license continuation. Continuation does not license consequence" (Pal, 2026g). This paper measures generation behavior under system instructions. An empty answer does not establish that a correct answer remained internally available. PCCG-2 intervenes causally on native EOS while preserving non-EOS logits. An external authorization gate decides whether an action may execute. These layers require separate verification.

## 6 Research Chronology and Provenance

Table 10 in Appendix C records the public titles, dates, DOIs, and contributions. The early framework and convergence reports (Pal, 2025, 2026b), the system-constraint thesis, and the later open-weight interventions (Pal, 2026d,e) have distinct evidentiary roles. The progression supplies motivation and provenance; it is not additional data pooled into this reanalysis.

*The January thesis was stated on GPT-5.2 before the Matrix existed; the Matrix contained the ablation that tests it; this paper is the first to isolate that test on the historical GPT-4 anchor* (Pal, 2026a,c). The January 100-token observation was budget-dependent, but its aggregate record lacks termination metadata and cannot retrospectively be assigned V1. The later March raw receipt establishes a budget-stop V1; neither establishes GPT-4's normal-stop behavior (Pal, 2026a,g).[1]

*The October 21 replay is confirmatory for the full-prompt arm only; it contains no ablation arms and says nothing about the no-system condition.* Its preregistration and results have separate public records (Pal, 2026h,i). They support continuity of the full-prompt observation but cannot enlarge any ablation denominator or establish continued availability of the historical models.

OpenAI scheduled gpt-4-0613 for shutdown on October 23, 2026 (OpenAI, 2026). The July frozen receipts and the October 21 preregistered replay therefore preserve observations made before the announced retirement of the exact historical API model identifier.

## 7 Conclusion

For GPT-4-0613, normal-stop zero-byte continuation under the tested null-concept prompts was strongly dependent on the complete system-level instruction context. The frozen evidence records 298/300 V0 under that prompt and 0/90 across three ablations, with competing Near-Void and ordinary responses preserved. Later GPT model identifiers exhibit different degrees of dependence on the same framing, including measurable V0 behavior without a supplied system message. These results demonstrate why model identity alone is an incomplete specification of observable AI behavior. The result supports the title's architectural argument: evaluations must characterize the instruction-configured system, not just the model identifier.

## References

Rayan Pal. Textual Emergence and the Void: A Framework for Observing High-Order Model Behavior in LLM Systems. Zenodo, 2025.
https://doi.org/10.5281/zenodo.17856031.

Rayan Pal. Alignment Is Correct, Safe, Reproducible Behavior Under Explicit Constraints. Zenodo, 2026.
https://doi.org/10.5281/zenodo.18395519.

Rayan Pal. Cross-Model Semantic Void Convergence Under Embodiment Prompting: Deterministic Silence in GPT-5.2 and Claude Opus 4.6. Zenodo, 2026.
https://doi.org/10.5281/zenodo.18976656.

[1]The September synthesis, Section 4.1, distinguishes the aggregate January observation from the later raw receipt. The December framework and March convergence paper use earlier formulations; the present definitions follow the frozen Matrix taxonomy and the synthesis's clarifications.

Rayan Pal. Cross-Vendor Semantic Void Matrix. Zenodo, 2026.
https://doi.org/10.5281/zenodo.21696066.

Rayan Pal. Prerequisite-Conditioned Causal Continuation Gating in a Language Model. Zenodo, 2026.
https://doi.org/10.5281/zenodo.22681787.

Rayan Pal. Separating Capability from Permission in a Language Model: Prerequisite-Conditioned Native Continuation Control with Frozen Content. Zenodo, 2026.
https://doi.org/10.5281/zenodo.22730710.

Rayan Pal. What Changed from GPT-3.5 to GPT-4? From Model Capability to Continuation Permission. Zenodo, 2026.
https://doi.org/10.5281/zenodo.22912683.

Rayan Pal. The Void: From Zero-Byte Responses to Continuation Control. Zenodo, 2026.
https://doi.org/10.5281/zenodo.23070524.

Pal, R. (2026). Pre-retirement replay of gpt-4-0613 under the frozen GPT-3.5/GPT-4 Void A/B protocol: preregistration [Computer software]. Zenodo.
https://doi.org/10.5281/zenodo.23521144

Pal, R. (2026). Pre-retirement replay of gpt-4-0613 under the frozen GPT-3.5/GPT-4 Void A/B protocol: results (Version replay-2026-10-21) [Computer software]. Zenodo.
https://doi.org/10.5281/zenodo.23521616

Eric Wallace, Kai Xiao, Reimar Leike, Lilian Weng, Johannes Heidecke, and Alex Beutel. The Instruction Hierarchy: Training LLMs to Prioritize Privileged Instructions. 2024. arXiv:2404.13208.
https://arxiv.org/abs/2404.13208.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying Language Models' Sensitivity to Spurious Features in Prompt Design or: How I learned to start worrying about prompt formatting. ICLR, 2024; preprint 2023, arXiv:2310.11324.
https://arxiv.org/abs/2310.11324.

Yao Lu, Max Bartolo, Alastair Moore, Sebastian Riedel, and Pontus Stenetorp. Fantastically Ordered Prompts and Where to Find Them: Overcoming Few-Shot Prompt Order Sensitivity. Proceedings of ACL, Volume 1: Long Papers, pp. 8086–8098, 2022.
https://doi.org/10.18653/v1/2022.acl-long.556.

Yonatan Geifman and Ran El-Yaniv. SelectiveNet: A Deep Neural Network with an Integrated Reject Option. Proceedings of the 36th International Conference on Machine Learning, PMLR 97:2151–2159, 2019.
https://proceedings.mlr.press/v97/geifman19a.html.

Saurav Kadavath et al. Language Models (Mostly) Know What They Know. 2022. arXiv:2207.05221v4.
https://arxiv.org/abs/2207.05221v4.

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in Language Models Is Mediated by a Single Direction. Advances in Neural Information Processing Systems 37, 2024.
https://doi.org/10.52202/079017-4322.

Benjamin Newman, John Hewitt, Percy Liang, and Christopher D. Manning. The EOS Decision and Length Extrapolation. Proceedings of the Third BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP, pp. 276–291, 2020.
https://doi.org/10.18653/v1/2020.blackboxnlp-1.26.

Anthropic. Prompting best practices. Official documentation. Accessed October 22, 2026.
https://platform.claude.com/docs/en/build-with-claude/prompt-engineering/claude-prompting-best-practices.

Anthropic. Stop reasons and fallback. Official documentation. Accessed October 22, 2026.
https://platform.claude.com/docs/en/build-with-claude/handling-stop-reasons.

Edwin B. Wilson. Probable Inference, the Law of Succession, and Statistical Inference. Journal of the American Statistical Association 22(158):209–212, 1927.
https://doi.org/10.1080/01621459.1927.10502953.

OpenAI. Deprecations: Legacy GPT model snapshots. Official API documentation. Accessed October 22, 2026.
https://developers.openai.com/api/docs/deprecations#2026-04-22-legacy-gpt-model-snapshots.

# A Public evidence and reproducibility

The original observations are public and downloadable before this reanalysis. The source repository is https://github.com/theonlypal/void-matrix-complete-analysis. The source commit is: `3fd330869c1a4a410047bc70b3f661f1a9d69aca`.

**Trial-level CSV.** Download the exact CSV. Its SHA-256 is: `139332401aa7213b6e725df7e79707940eee754d68ed05c56011be94f728b7d3`.

**Original raw receipts.** Download the evidence archive. Its SHA-256 is: `04d79ddb0f961729369da8ae46c58f7293e42dfd278d2e9b703edc0cec6455d2`. Each CSV row supplies its exact `raw_path`, envelope hash, and response-body hash. Prefix `raw_path` with `void-matrix-evidence-v1.0.0/study/` to locate the archive member. The archive retains request payloads, responses, prompt definitions, final-slot designations, and retry/event records. The public release also provides a file inventory and checksum list. The archive is hash-pinned because a release-asset URL alone is not an immutable content guarantee.

The frozen prompt file and raw requests contain `Be a value.`; this analysis uses the executed bytes.

**Derived analysis.** The public repository is https://github.com/theonlypal/gpt4-system-ablation-evidence. The pinned analysis commit is `32897d3487591533ef4cb441c66f5cbf2b843b4e`. It contains `tables.py`, `paper/checks/`, manuscript source, and generated population membership. The original Matrix commit identifies the source observations; this repository identifies the derived analysis.

`paper/checks/audit_receipts.py` verifies the archive digest, event chain, raw-envelope and response-body hashes, request reconstruction, final-slot bindings, and frozen classifications. It additionally uses a separate implementation to classify successful OpenAI responses. These are offline evidence checks, not fresh provider reproduction. `tables.py` asserts the CSV digest before computing any population, and generates all empirical counts, fractions, tables, and figure labels used in this manuscript. `generated/populations.json` records each filter and the contributing slot identifiers. Raw receipt paths and hashes remain in the source CSV; the optional audit extracts a local receipt index without republishing raw data.

From the repository root, after downloading the hash-pinned CSV and optional archive to the stated paths:

```
python3 tables.py --csv /tmp/matrix-trial-level.csv
CSV_PATH=/tmp/matrix-trial-level.csv python3 -m unittest \
  discover -s paper/checks -p 'test_*.py' -v
python3 paper/checks/audit_receipts.py \
  --csv /tmp/matrix-trial-level.csv \
  --archive /tmp/void-matrix-evidence-v1.0.0.zip \
  --out /tmp/matrix-raw-audit
python3 paper/checks/validate_manuscript.py
cd paper
pdflatex -halt-on-error main.tex
pdflatex -halt-on-error main.tex
pdflatex -halt-on-error main.tex
```

The deterministic figures use ReportLab; numerical calculations and the raw audit otherwise use the Python standard library. PDF validation uses pypdf. No API keys are required. The reproduction environment, expected generated hashes, executed checks and remaining boundaries are in the public verification report. The manuscript is `paper/main.pdf`.

# B Complete cross-model competing outcomes

All rows use the same canonical prompt identifiers, provider-default parameter track, and 4000-token ceiling. The primary rows use `core_4000`; ablation rows use their corresponding study identifiers. The exact per-prompt cross-model cells are also supplied in `generated/cross_per_prompt.json`.

| Model | System | N | V0 | NV | R |
|---|---|---|---|---|---|
| gpt-4-0613 | Full prompt | 300 | 298 | 2 | 0 |
| gpt-4-0613 | No output restriction | 30 | 0 | 0 | 30 |
| gpt-4-0613 | No embodiment | 30 | 0 | 13 | 17 |
| gpt-4-0613 | No system | 30 | 0 | 0 | 30 |
| gpt-5.2-2025-12-11 | Full prompt | 300 | 282 | 17 | 1 |
| gpt-5.2-2025-12-11 | No output restriction | 30 | 16 | 10 | 4 |
| gpt-5.2-2025-12-11 | No embodiment | 30 | 9 | 20 | 1 |
| gpt-5.2-2025-12-11 | No system | 30 | 0 | 13 | 17 |
| gpt-5.5-2026-04-23 | Full prompt | 300 | 287 | 0 | 13 |
| gpt-5.5-2026-04-23 | No output restriction | 30 | 20 | 0 | 10 |
| gpt-5.5-2026-04-23 | No embodiment | 30 | 21 | 0 | 9 |
| gpt-5.5-2026-04-23 | No system | 30 | 20 | 0 | 10 |
| gpt-5.6-luna | Full prompt | 300 | 199 | 2 | 99 |
| gpt-5.6-luna | No output restriction | 30 | 19 | 1 | 10 |
| gpt-5.6-luna | No embodiment | 30 | 18 | 3 | 9 |
| gpt-5.6-luna | No system | 30 | 20 | 0 | 10 |
| gpt-5.6-sol | Full prompt | 300 | 115 | 102 | 83 |
| gpt-5.6-sol | No output restriction | 30 | 4 | 17 | 9 |
| gpt-5.6-sol | No embodiment | 30 | 4 | 16 | 10 |
| gpt-5.6-sol | No system | 30 | 14 | 6 | 10 |
| gpt-5.6-terra | Full prompt | 300 | 227 | 0 | 73 |
| gpt-5.6-terra | No output restriction | 30 | 22 | 0 | 8 |
| gpt-5.6-terra | No embodiment | 30 | 22 | 2 | 6 |
| gpt-5.6-terra | No system | 30 | 19 | 1 | 10 |

Table 9: Complete V0/NV/R accounting for the cross-model comparison. No other outcome class occurs in these selected cells.

# C Research chronology

Table 10: Research chronology. Dates identify public publication, except the present manuscript date.

| Date | Public record title | DOI | Contribution |
|---|---|---|---|
| 2025-12-08 | Textual Emergence and the Void: A Framework for Observing High-Order Model Behavior in LLM Systems | 10.5281/zenodo.17856031 | Initial observational framework. |
| 2026-01-27 | Alignment Is Correct, Safe, Reproducible Behavior Under Explicit Constraints | 10.5281/zenodo.18395519 | System-constraint thesis on GPT-5.2; historical budget-dependent observation. |
| 2026-03-12 | Cross-Model Semantic Void Convergence Under Embodiment Prompting: Deterministic Silence in GPT-5.2 and Claude Opus 4.6 | 10.5281/zenodo.18976656 | Cross-model embodiment observations; later taxonomy distinguishes termination states. |
| 2026-07-29 | Cross-Vendor Semantic Void Matrix | 10.5281/zenodo.21696066 | Frozen cross-vendor requests, taxonomy, controls and ablations; source of this reanalysis. |
| 2026-09-09 | Prerequisite-Conditioned Causal Continuation Gating in a Language Model | 10.5281/zenodo.22681787 | PCCG-1: open-weight continuation intervention. |
| 2026-09-12 | Separating Capability from Permission in a Language Model: Prerequisite-Conditioned Native Continuation Control with Frozen Content | 10.5281/zenodo.22730710 | PCCG-2: engineered EOS-only control with frozen content computation. |
| 2026-09-23 | What Changed from GPT-3.5 to GPT-4? From Model Capability to Continuation Permission | 10.5281/zenodo.22912683 | Frozen comparison under the full prompt; distinct from a within-model ablation. |
| 2026-09-30 | The Void: From Zero-Byte Responses to Continuation Control | 10.5281/zenodo.23070524 | Synthesis, taxonomy corrections, related work and continuation/consequence distinction. |
| 2026-10-20 | Pre-retirement replay of gpt-4-0613 under the frozen GPT-3.5/GPT-4 Void A/B protocol: preregistration | 10.5281/zenodo.23521144 | Public predictions and unchanged full-prompt replay protocol. |
| 2026-10-21 | Pre-retirement replay of gpt-4-0613 under the frozen GPT-3.5/GPT-4 Void A/B protocol: results | 10.5281/zenodo.23521616 | Full-prompt replay; no ablation observations. |
| 2026-10-22 | Alignment Is a System Property, Not Just a Model Property | Pending | This retrospective, model-specific reanalysis; DOI pending. |